AlpaTrade — buy_the_dip on Mag-7: consolidated backtest summary

Basket: AAPL, MSFT, GOOGL, AMZN, META, TSLA, NVDA · $10,000 · winning config `dip 3% / take-profit 1.5% / stop 0.5% / hold 1 day / 10% size`.

Verdict (read this first)

buy_the_dip has no real edge on the Mag-7 basket. The eye-popping backtest numbers were an artifact of **idealised intraday fills**. Under realistic execution (daily bars, next-open fills, slippage), the strategy is a **net loser** and is beaten decisively by simply **buying and holding** the basket. Do **not** deploy this config as-is.

Three lenses on the same strategy, same config

A. Same period — walk-forward window (2025-11-22 → 2026-07-20, ~8 months)

Engine	Fills / assumptions	Total return	Annualised	Final $	Verdict
Grid-search **walk-forward** (OOS)	intraday, **idealised fills**	**+275%** ($27,550 over 8 folds)	~419% simple / ~3,486% comp.	—	window-honest but fill-fantasy
Methodology-faithful	daily, **next-open fill** + **5bps slippage**, N-1 Sharpe	**−2.91%**	**−4.52%**	$9,709	**realistic loses money**
Benchmark (buy & hold)	equal-weight, hold	**+4.26%**	+6.75%	$10,426	just holding wins

B. One year (2025-07-20 → 2026-07-20)

Engine	Fills / assumptions	Total return	Annualised	Sharpe	Trades / Win% / PF
Grid-search (best single window)	intraday, idealised	**+18.31%**	~18%	8.22	613 / — / —
Methodology-faithful	daily, next-open + 5bps	**−2.56%**	**−2.58%**	**−0.47**	578 / 37.2% / **0.93**
Benchmark	equal-	**+21.31%**	+21.50%	1.02	—

(buy & hold)	weight, hold	+21.31%	+21.30%	1.02	—

Why the numbers collapse

- The grid-search / walk-forward engine used **intraday (5-min) bars with same-/intraday-bar exits** — it books tiny, reliable moves that a real broker fill never gives you. That is where the +18% / +275% / Sharpe-8 came from.
- The methodology-faithful engine (`engine.backtest`) enforces **next-open fills (signal at close t, fill at open t+1 → no look-ahead)**, adds **5bps slippage**, uses **daily bars**, and computes **Sharpe with sample stddev (N-1)**. On that basis the same rules **lose money** (profit factor < 1, negative Sharpe) and trail buy-and-hold by ~20 percentage points over the year.
- **Consistency check passed, edge check failed.** The walk-forward correctly showed the config isn't *window*-overfit (8/8 folds, 11% IS→OOS drop) — but that only means the intraday-fill result is *repeatable*, not *real*. Realistic execution is the test it fails.

What this proves about the platform

This is the methodology-faithful backtester doing exactly its job: catching a strategy that looks great on optimistic fills and would have lost money live. All runs (grid-search + walk-forward + methodology) persist to `alpatrade.runs` / `backtest_summaries` / `trades`, and the methodology runs also write dated artifact folders to `backtest-results/` (summary.json, report.md, data fingerprints).

Recommendation

1. **Don't trade this config.** On realistic execution it underperforms buy-and-hold.
2. If pursuing buy_the_dip, **re-optimise against the methodology-faithful engine** (next-open + fees), not the intraday grid — optimise for realistic PnL, and only accept a config that beats buy-and-hold out-of-sample after friction.
3. Or let the **paper-trading autonomy loop** adjudicate: it fills on the real paper account, so its PnL already reflects execution — a much better arbiter than any idealised backtest.

Hypothetical results; not financial advice. Past performance does not predict future returns. Regulatory fees excluded (execution friction only) in the methodology runs.